

1. Administrative & Study Identification

Full Study Title: A Phase 1/2 Randomized, Umbrella Study to Evaluate the Efficacy and Safety of MK-2870 Plus Enfortumab Vedotin (EV) With and Without Pembrolizumab, as Treatment for Participants With Advanced Urothelial Carcinoma (KEYMAKER-U04): Substudy 04C **Short Title/Acronym:** MK-3475-04C / KEYMAKER-U04 Substudy 04C / A Study of Efficacy and Safety of Sacituzumab Tirumotecan (MK-2870) Plus Enfortumab Vedotin (EV) With and Without Pembrolizumab in Advanced Urothelial Carcinoma **Protocol Number:** MK-3475-04C (Secondary IDs: 3475-04C, 2023-506387-14) **NCT Number:** NCT06483334 **Sponsor Name:** Merck (Merck Sharp & Dohme LLC) **Investigational Product:** Sacituzumab Tirumotecan (MK-2870) + Enfortumab Vedotin (EV) \$\\pm\$ Pembrolizumab **Phase of Development:** Phase 1 / Phase 2 **Trial Status:** Recruiting **Medical Monitor:** Merck Sharp & Dohme Clinical Operations Lead (Contact: 1-855-702-8222)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety profile in Part 1 by determining the percentage of participants with dose-limiting toxicities (DLTs), experiencing at least one adverse event (AE), and discontinuing study treatment due to an AE. **Secondary Objectives:** To evaluate the preliminary efficacy, specifically Objective Response Rate (ORR) in Part 1 and Duration of Response (DOR) in Part 2, and to evaluate the pharmacokinetics, specifically the Maximum Serum Concentration (Cmax) of Sacituzumab Tirumotecan-ADC in Part 1.

2.2 Background & Rationale To address the clinical need for effective combination therapies in patients with locally advanced/metastatic urothelial carcinoma (la/mUC) who have previously received or are ineligible for standard treatments. Combining an anti-TROP2 antibody-drug conjugate (ADC) (sacituzumab tirumotecan) with an anti-nectin-4 ADC (enfortumab vedotin), and evaluating the addition of an immune checkpoint inhibitor (pembrolizumab), aims to optimize tumor response and long-term clinical outcomes in this patient population. The substudy consists of 2 parts: Part 1 evaluating safety/efficacy of the ADC doublet, and Part 2 (based on Part 1 results) evaluating the addition of pembrolizumab.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Umbrella Substudy) Control Type: Parallel Assignment Blinding: Open-label Randomization: Randomized

3.2 Planned Enrollment & Duration Target N\$: 98 participants Number of Sites: Multiple global sites Estimated Study Start: 2024 Estimated Primary Completion: ~2029 (Expected to report results ~3 years from current date)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Must have histologically documented, locally advanced/metastatic urothelial carcinoma (la/mUC). 2. Must provide an archival tumor tissue sample or newly obtained core or excisional biopsy of a tumor lesion demonstrating UC, not previously irradiated, and adequate for biomarker evaluation. A newly obtained biopsy is strongly preferred, but not required if archival tissue is evaluable. 3. Any AEs due to previous anticancer therapies must have recovered to $\leq$ Grade 1 or baseline. Endocrine-related AEs adequately treated with hormone replacement are eligible. 4. **PART 1 ONLY:** Participants must have received platinum-based chemotherapy for treatment of la/mUC and must not have received >2 lines of therapy for la/mUC. Platinum-based chemotherapy followed by avelumab maintenance is considered 2 lines of therapy. 5. **PART 2 ONLY:** Participants must not have received prior systemic therapy for la/mUC.

4.2 Exclusion Criteria 1. Known additional malignancy that is progressing or has required active treatment within the past 3 years. 2. Known active central nervous system metastases and/or carcinomatous meningitis. 3. Has Grade ≥ 2 peripheral neuropathy. 4. Has history of documented severe dry eye syndrome, severe Meibomian gland disease and/or blepharitis, or corneal disease that prevents/delays corneal healing. 5. Has active inflammatory bowel disease requiring immunosuppressive medication or previous history of inflammatory bowel disease (e.g., Crohn's disease, ulcerative colitis, or chronic diarrhea). 6. Has uncontrolled, significant cardiovascular disease or cerebrovascular disease and/or serious cardiovascular and cerebrovascular diseases within the 6 months preceding study intervention. 7. Has active keratitis or corneal ulcerations. Superficial punctate keratitis is allowed if the disorder is being adequately treated in the opinion of the investigator. 8. Has a history of uncontrolled diabetes. 9. Has received an investigational agent or has used an investigational device within 4 weeks prior to study intervention administration. 10. Has received a live or live-attenuated vaccine within 30 days prior to the first dose of study intervention. 11. **PART 2 ONLY:** Has a diagnosis of immunodeficiency or is receiving chronic systemic steroid therapy or any other form of immunosuppressive

therapy within 7 days before the first dose of study intervention. Inhaled or topical steroids are permitted in the absence of active autoimmune disease. Physiologic replacement doses of corticosteroids are permitted for participants with adrenal insufficiency. 12. **PART 2 ONLY:** Has an active autoimmune disease that has required systemic treatment in past 2 years except replacement therapy. 13. Is human immunodeficiency virus (HIV)-infected and has a history of Kaposi's sarcoma and/or Multicentric Castleman's Disease. 14. Has active Hepatitis B or Hepatitis C virus infection. 15. Has a history of (noninfectious) pneumonitis/interstitial lung disease that required steroids or has current pneumonitis/interstitial lung disease. 16. Has an active infection requiring systemic therapy. 17. **PART 2 ONLY:** History of allogeneic tissue/solid organ transplant. 18. Has not adequately recovered from major surgery or has ongoing surgical complications.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Participants receive Sacituzumab Tirumotecan and Enfortumab Vedotin (EV) on Days 1 and 8 of every 3-week cycle. Participants in the Pembrolizumab arm receive 200 mg of Pembrolizumab on Day 1 of every 3-week cycle. **Route of Administration:** Intravenous (IV) Infusion for all study drugs. **Duration of Treatment:** Administered until disease progression, intolerable toxicity, or investigator decision. Pembrolizumab administration is capped at approximately 2 years (35 cycles).

5.2 Concomitant & Prohibited Therapy Permitted: Physiologic replacement doses of corticosteroids for adrenal insufficiency; inhaled or topical steroids in the absence of active autoimmune disease. Treatment for superficial punctate keratitis is allowed. **Prohibited:** Any investigational agent within 4 weeks prior to study intervention; live or live-attenuated vaccines within 30 days prior to the first dose; systemic immunosuppressive therapy or chronic systemic steroid therapy within 7 days prior to the first dose (for Part 2).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Biopsy/Archival Tissue Collection, Safety Labs
Baseline	Day 1	Randomization, First IV Infusion (Sacituzumab Tirumotecan + EV + Pembrolizumab)
Treatment Cycles	Every 3 Weeks	Day 1 & 8: IV Infusion of ADCs; Day 1: Pembrolizumab (if applicable). Efficacy/Safety monitoring.
End of Study	Up to ~2 Years	Final Efficacy Assessment, Safety Labs, Exit Interview, Disease Progression Tracking

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints * **Part 1: Percentage of Participants with Dose-limiting toxicities (DLT):** A DLT is defined as an event with toxicity including the type, severity, time of onset, time of resolution, and the probable association with study treatment that are not due to pre-existing conditions as defined by the NCI CTCAE Version 5.0. * **Part 1: Percentage of Participants Who Experienced At Least One Adverse Event (AE):** An AE is defined as any untoward medical occurrence in a participant administered study drug, which does not necessarily have to have a causal relationship with the study drug. * **Part 1: Percentage of Participants Who Discontinued Study Treatment Due to an AE:** Discontinuation rate due to any untoward medical occurrence as defined above.

7.2 Secondary Endpoints * **Part 1: Objective Response Rate (ORR):** Assessed per response criteria to evaluate preliminary efficacy. * **Part 2: Duration of Response (DOR):** Time from the first documented evidence of response until progressive disease (PD) or death. * **Part 1: Maximum Serum Concentration (Cmax) of Sacituzumab Tirumotecan-Antibody-Drug Conjugate (ADC):** Evaluation of pharmacokinetics.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous site monitoring for standard systemic, dermatologic, and neurologic AEs (e.g., peripheral neuropathy assessment) typical of ADCs. **Serious Adverse Events (SAEs):** Standard reporting timeline (typically 24 hours of site awareness). **Data Monitoring Committee (DMC):** Internal/External safety review committees utilized for dose-escalation and cohort safety gating in umbrella master studies.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for efficacy outcomes; Safety Set for adverse event profiling. **Sample Size Justification:** Target N = 98; powered adequately for preliminary phase 1/2 signal detection of safety, DLT incidence, and efficacy profile evaluation. **Handling of Missing Data:** Participants with missing data or no post-baseline tumor assessments are typically handled as non-responders in efficacy calculations per protocol rules.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Scheduled onsite/remote monitoring visits to track radiographic reading compliance and safety parameters.
Audit Trail: Standard EDC (Electronic Data Capture) systems with full audit trails and eCRF locking mechanisms.

11. Study Identifiers & Governance

Primary ID: {{MK-3475-04C}} **Registry Number:** {{NCT06483334}}
Sponsor/Lead Organization: {{Merck}} **Study Title:** {{KEYMAKER-U04: Substudy 04C}} **Official Regulatory Title:** {{A Phase 1/2 Randomized, Umbrella Study to Evaluate the Efficacy and Safety of MK-2870 Plus Enfortumab Vedotin (EV) With and Without Pembrolizumab, as Treatment for Participants With Advanced Urothelial Carcinoma}}

12. Clinical Framework (Urothelial Carcinoma Specific)

Primary Condition: {{Metastatic Urothelial Carcinoma, Locally Advanced Urothelial Carcinoma}} **Diagnosis Threshold:** {{Histologically Documented UC with target measurable lesions}}
Medical Methodology: {{Combination Antibody-Drug Conjugates (ADC) & Immune Checkpoint Inhibition}} **Optimization Target:** {{Tolerability/Safety, Objective Response Rate (ORR), and Duration of Response (DOR)}}

13. Study Procedures & Methodology

Management Pathway: {{Oncology infusion pathway for ADCs + Immunotherapy}} **Education Protocol (HCP):** {{Management of ADC-specific toxicities including peripheral neuropathy and hyperglycemia}} **Education Protocol (Patient):** {{Onsite infusion education and symptom reporting for immunological/ADC side effects}} **Quality Audit Mechanism:** {{Central and local radiologic tumor assessment tracking}}

14. Observation & Follow-Up Schedule

Total Duration: {{Treatment up to ~2 years (35 cycles) + Progression Follow-up}} **Onsite Visit Cadence:** {{Days 1 and 8 of every 3-week cycle}} **Remote Monitoring Frequency:** {{As clinically indicated between cycles}} **Checklist Review Interval:** {{Every tumor assessment milestone (typically every 6-9 weeks)}}

15. Sign-off & Approval

| Role Name | Signature | Date | | :--- | :--- | :--- | |
Principal Investigator | ____ | [DD-MMM-YYYY] | | **Clinical**

Operations Lead | ____ | [DD-MMM-YYYY] | | Lead Statistician |
____ | [DD-MMM-YYYY] |